u_link

versus_link

clk
frame_tick

got_attack

got_garbage

got_lost got_start

link_ok

rx
send_attack
send_garbage

send_lost send_start

tx

attack_id[4:0]

garbage_n[4:0] got_attack_id[4:0]

got_n[4:0]

my_height[4:0]

opp_height[4:0]

att_id_reg[4:0]

RTL_REG C

CED Q

att_p_i RTL_MUX

I0 S=1'b1

I1 S=default O S

att_p_i__0 RTL_MUX

I0 S=1'b1

I1 S=default O S

att_p_i__1 RTL_MUX

I0 S=1'b1

I1 S=default O S att_p_i__2 RTL_MUX

I0 S=1'b1

I1 S=default O S

att_p_i__3 RTL_MUX

I0 S=1'b1

I1 S=default O S

att_p_i__4 RTL_MUX

I0 S=1'b1

I1 S=default O S

att_p_i__5 RTL_MUX

I0 S=1'b1

I1 S=default O S

att_p_i__6 RTL_MUX

I0 S=1'b1

I1 S=default O S

att_p_reg RTL_REG C

CED Q

g_att_i RTL_ROMO A[2:0]

g_att_i__0 RTL_MUX

I0 S=1'b1

I1 S=default O S

g_att_id_i RTL_ROMO A[2:0] g_att_id_i__0 RTL_MUX

I0 S=1'b1

I1 S=default O S g_att_id_reg[4:0]

RTL_REG C

CED Q

g_att_reg RTL_REG_SYNC CD Q

RST

g_garb_i RTL_ROMO A[2:0]

g_garb_i__0 RTL_MUX

I0 S=1'b1

I1 S=default O S

g_garb_reg RTL_REG_SYNC CD Q

RST

g_lost_i RTL_ROMO A[2:0]

g_lost_i__0 RTL_MUX

I0 S=1'b1

I1 S=default O S

g_lost_reg RTL_REG_SYNC CD Q

RST

g_n_i RTL_ROMO A[2:0]

g_n_i__0 RTL_MUX

I0 S=1'b1

I1 S=default O S g_n_reg[4:0]

RTL_REG C

CED Q

g_start_i RTL_ROMO A[2:0]

g_start_i__0 RTL_MUX

I0 S=1'b1

I1 S=default O S

g_start_reg RTL_REG_SYNC CD Q

RST

garb_p_reg[4:0] RTL_REG_SYNC CD Q

SET

gsum_i RTL_MUX S

I0[4:0] S=1'b1

I1[4:0] S=default O[4:0]

gsum_i__0 RTL_MUX S

I0[4:0] S=1'b1

I1[4:0] S=default O[4:0] gsum_i__1 RTL_MUX S

I0[4:0] S=1'b1

I1[4:0] S=default O[4:0] gsum_i__2 RTL_MUX S

I0[4:0] S=1'b1

I1[4:0] S=default O[4:0] gsum_i__3 RTL_MUX S

I0[4:0] S=1'b1

I1[4:0] S=default O[4:0]

gsum_i__4 RTL_MUX S

I0[4:0] S=1'b1

I1[4:0] S=default O[4:0]

gsum_i__5 RTL_MUX

I0 S=1'b1

I1 S=default O S

gsum0_i +RTL_ADD

I0[4:0]

I1[4:0] O[5:0]

gsum0_i__0 >RTL_GT O

I0[5:0]
I1[5:0] V=B"011111"

hb_cnt_i RTL_MUX

I0[2:0] S=3'b101

I1[2:0] S=default O[2:0] S[2:0]

hb_cnt_reg[2:0]

RTL_REG C

CED Q hb_cnt0_i +RTL_ADD

I1

I0[2:0] O[2:0]

hb_p_i RTL_MUX

I0 S=1'b1

I1 S=default O S hb_p_i__0 RTL_MUX

I0 S=1'b1

I1 S=default O S hb_p_i__1 RTL_MUX

I0 S=1'b1

I1 S=default O S

hb_p_i__2 RTL_MUX

I0 S=1'b1

I1 S=default O S hb_p_i__3 RTL_MUX

I0 S=1'b1

I1 S=default O S hb_p_i__4 RTL_MUX

I0 S=1'b1

I1 S=default O S

hb_p_i__5 RTL_ROMO A[2:0] hb_p_i__6 RTL_MUX

I0 S=1'b1

I1 S=default O S

hb_p_i__7 RTL_MUX

I0 S=1'b1

I1 S=default O S hb_p_i__8 RTL_MUX

I0 S=1'b1

I1 S=default O S hb_p_i__9 RTL_MUX

I0 S=1'b1

I1 S=default O S

hb_p_i__10 RTL_MUX

I0 S=1'b1

I1 S=default O S hb_p_i__11 RTL_MUX

I0 S=1'b1

I1 S=default O S hb_p_i__12 RTL_MUX

I0 S=1'b1

I1 S=default O S

hb_p_reg RTL_REG C

CED Q

lost_p_i RTL_MUX

I0 S=1'b1

I1 S=default O S

lost_p_i__0 RTL_MUX

I0 S=1'b1

I1 S=default O S

lost_p_i__1 RTL_MUX

I0 S=1'b1

I1 S=default O S

lost_p_i__2 RTL_MUX

I0 S=1'b1

I1 S=default O S

lost_p_i__3 RTL_MUX

I0 S=1'b1

I1 S=default O S

lost_p_i__4 RTL_MUX

I0 S=1'b1

I1 S=default O S
lost_p_reg RTL_REG C

CED Q

opp_h_i

RTL_MUX

I0 S=3'b000
I1 S=3'b001
I2 S=3'b010
I3 S=3'b011
I4 S=3'b100

I5 S=default O
S[2:0]

opp_h_i__0 RTL_MUX

I0 S=1'b1

I1 S=default O S

opp_h_reg[4:0]

RTL_REG C

CED Q

opp_h0_i <=RTL_LEQ O

I0[4:0]
I1[4:0] V=B"10100"

silence_reg[4:0]

RTL_REG_SYNC C

CED Q

RST

silence0_i +RTL_ADD

I1

I0[4:0] O[4:0]

silence0_i__0 RTL_AND

I0

I1 O

silence1_i <RTL_LT O

I0[5:0]
I1[5:0] V=B"011110"

start_p_i RTL_MUX

I0 S=1'b1

I1 S=default O S

start_p_i__0 RTL_MUX

I0 S=1'b1

I1 S=default O S

start_p_i__1 RTL_MUX

I0 S=1'b1

I1 S=default O S

start_p_i__2 RTL_MUX

I0 S=1'b1

I1 S=default O S

start_p_reg RTL_REG C

CED Q

tx_data_i RTL_MUX S

I0[7:0] S=1'b1

I1[7:0] S=default O[7:0]

tx_data_i__0 RTL_MUX S

I0[7:0] S=1'b1

I1[7:0] S=default O[7:0]

tx_data_i__1 RTL_MUX S

I0[7:0] V=B"01000000", S=1'b1

I1[7:0] S=default O[7:0]

tx_data_i__2 RTL_MUX S

I0[7:0] V=B"01100000", S=1'b1

I1[7:0] S=default O[7:0]

tx_data_i__3 RTL_MUX

I0 S=1'b1

I1 S=default O S

tx_data_i__4 RTL_MUX

I0 S=1'b1

I1 S=default O S

tx_data_i__5 RTL_MUX

I0 S=1'b1

I1 S=default O S

tx_data_i__6 RTL_MUX

I0 S=1'b1

I1 S=default O S

tx_data_i__7 RTL_MUX

I0 S=1'b1

I1 S=default O S

tx_data_reg[7:0]

RTL_REG C

CED Q

tx_start_i RTL_MUX

I0 S=1'b1

I1 S=default O S

tx_start_i__0 RTL_MUX

I0 S=1'b1

I1 S=default O S

tx_start_i__1 RTL_MUX

I0 S=1'b1

I1 S=default O S

tx_start_i__2 RTL_MUX

I0 S=1'b1

I1 S=default O S

tx_start_i__3 RTL_MUX

I0 S=1'b1

I1 S=default O S

tx_start_reg RTL_REG_SYNC CD Q

RST SET
tx_start0_i >RTL_GT O

I0[4:0]
I1[4:0]

tx_start1_i RTL_AND

I0

I1 O

tx_start2_i =RTL_EQ

I0

I1 O

tx_start2_i__0 =RTL_EQ

I0

I1 O

u_rx uart_rx

clk

rx data[7:0] valid

u_tx

uart_tx

busy clk

start

tx data[7:0]